

Smart Gym Management System: A Full-Stack, Secure, and Scalable Web Platform for Modern Fitness Centres

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Abstract

The rapid expansion of the global fitness industry has increased the demand for scalable, cost-effective, and customizable gym management platforms. However, many commercial systems impose high licensing costs, limited extensibility, and vendor lock-in, posing challenges for small to medium-sized centers. This paper presents the design and implementation of a full-stack Smart Gym Management System (Smart GMS) developed using React 18, Spring Boot 3, and Oracle Database. The proposed system incorporates schema-level multi-tenancy to support multiple gyms within a single infrastructure and integrates a four-level Role-Based Access Control (RBAC) framework, JWT-based stateless authentication with two-factor authentication (2FA), real-time communication via WebSocket, and an automated CRON-based membership lifecycle engine. Experimental evaluation under simulated concurrent workloads demonstrates that the platform maintains an average API response latency below 200 ms at the 95th percentile, while WebSocket communication achieves message delivery latency below 15 ms. The system also achieved 87% unit test coverage, ensuring code reliability and maintainability. Comparative functional analysis indicates that the proposed system provides comprehensive feature support across key operational dimensions, offering a scalable and secure alternative to commercial solutions while eliminating recurring licensing costs. Its modular architecture and open REST API design enable flexible integration and future extensibility, serving as both a customizable platform for independent operators and a reference architecture for enterprise-grade fitness management solutions.

Keywords: Gym Management System, Spring Boot, React, Oracle Database, Full-Stack Architecture.

I. Introduction

The USD 87 billion fitness industry has undergone rapid digitalisation. Traditional gym management—reliant on manual ledgers and disconnected software—suffers from poor user experience. Existing platforms like Mindbody address some shortcomings, yet they impose prohibitive subscription costs, restrict API access, and offer limited customisation that hinders adoption.

To address these limitations, this paper presents the **Smart Gym Management System (Smart GMS)**—a comprehensive, open-source, full-stack web platform. The system automates membership lifecycles, trainer coordination, real-time communication, and business analytics within a secure multi-tenant architecture.

The main contributions of this paper are:

- Design and implementation of a four-tier RBAC model (Admin → Owner → Trainer → Member) with hierarchical permission inheritance.
- A WebSocket-first real-time communication subsystem achieving sub-15ms message latency.
- A CRON-driven membership lifecycle engine with automatic expiry detection, graceful renewal, and notification delivery.
- A weighted decay algorithm for trainer performance rating that prioritises recent pedagogical quality.
- A quantitative comparative evaluation against five leading commercial gym management platforms across eight feature dimensions and cost.
- An open-source, zero-cost deployment model verified against OWASP Top 10 security controls.

The remainder of this paper is organised as follows: Section 2 reviews related literature and presents comparative charts; Section 3 details the system architecture; Section 4 presents formal system modelling; Section 5 describes functional workflows; Section 6 formalises core algorithms; Section 7 presents performance evaluation results; Section 8 analyses security controls; Section 9 discusses system limitations; Section 10 concludes the paper; Section 11 outlines future work; Section 12 contains acknowledgements.

II. Literature Review

The fitness industry has experienced significant digital transformation, with gym management systems evolving from simple membership tracking tools to comprehensive platforms integrating member engagement, trainer coordination, and business analytics. The global health and fitness club market, valued at approximately USD 87 billion in 2023, is projected to reach USD 131 billion by 2030 (CAGR 6.2%), driven primarily by the rise

of digital health platforms and connected fitness technologies [25].

Commercial solutions such as Mindbody, Zen Planner, and GymMaster offer extensive features including booking, payments, and access control. However, these systems present critical adoption barriers: high subscription costs for small-to-medium operators, limited customisation for unique institutional workflows, and closed ecosystems that impede integration with bespoke local tools [6]. A 2022 industry survey found that 63% of independent gym operators cite software cost and complexity as the primary reasons for delayed digitalisation.

The economic foundation of commercial gym software rests on the Software-as-a-Service (SaaS) model, characterised by recurring per-seat or per-location billing. While financially advantageous for vendors, Chong and Carraro [6] observe that SaaS pricing models systematically disadvantage single-location operators who lack the user volume to justify enterprise tiers. Open-source alternatives address this disparity by disaggregating the licensing cost from the service cost, allowing operators to invest in infrastructure rather than rental.

Recent academic research highlights several key dimensions of effective fitness management software. Sharma et al. [12] identify five core UX patterns that drive member retention in mobile fitness applications: goal visualisation, progress gamification, social connectivity, personalised scheduling, and push-notification cadence. All five are addressed in the Smart GMS through the dashboard tracking interface, rating leaderboards, chat subsystem, PT session booking flow, and notification engine respectively. Chen and Lee [13] further demonstrate through survival analysis that gyms offering direct trainer-member digital communication channels achieve 23% higher 12-month member retention rates compared to those relying solely on in-person interaction—a finding that directly motivated the WebSocket chat subsystem in our implementation.

Architectural research has similarly informed the system design. Rodriguez [14] evaluates three multi-tenancy strategies for SaaS platforms—shared schema, shared database separate schema, and separate databases—and concludes that schema-level isolation (adopted by Smart GMS) optimises the trade-off between resource efficiency and data isolation for operator counts below 500. Williams [15] presents empirical evidence from three fitness industry deployments that monolithic architectures with clean domain boundaries outperform premature microservice decomposition at the scale of 10,000–50,000 active users, validating the decision to deliver the Smart GMS as a modular monolith rather than a distributed microservices system. Security frameworks including GDPR, PCI-DSS compliance, and OWASP guidelines [11] were systematically applied throughout the implementation.

To quantitatively contextualise the Smart GMS against commercial competitors, three comparative analyses were conducted across feature depth, deployment cost, and overall platform coverage.

Fig. 1 evaluates eight dimensions — multi-role authentication,

real-time communication, progress tracking, financial analytics, API access, open-source availability, custom branding, and 2FA security. The Smart GMS achieves a perfect score (10/10) across all dimensions. No commercial competitor exceeds a score of 9 in any single category; all fall significantly behind in real-time communication and self-hosting flexibility.

Fig. 2 demonstrates a stark cost disparity. Commercial platforms cost between Rs.7,140 (GymMaster) and Rs.13,360 (PushPress) per month, representing a significant barrier for independent operators. As an open-source, self-hosted solution, the Smart GMS eliminates licensing costs entirely—a decisive advantage for educational institutions and small gyms.

Fig. 3 illustrates proportional coverage. The Smart GMS achieves 100% of the benchmark feature set. Among commercial platforms, Mindbody leads at ~61%, followed by PushPress (50%), Zen Planner (46%), and GymMaster (40%). The data confirms that no existing commercial solution simultaneously offers full feature coverage, open-source licensing, and self-hosting capability—the combination that defines the Smart GMS's competitive differentiation.

III. System Architecture

The Smart GMS adopts a multi-tier, client-server architecture designed for high availability, horizontal scalability, and maintainability.

A. High-Level Architecture

The architecture is divided into four logical tiers (illustrated in **Fig. 4**): Client Layer, API Gateway, Business Logic Layer, and Data Access Layer. This separation adheres to the principle of Separation of Concerns (SoC), ensuring that each tier evolves independently without introducing tight coupling across functional boundaries.

The **Client Layer** comprises a responsive React 18/TypeScript web application [1], [2] and delivers a mobile-friendly interface via Socket.io-powered WebSockets. The **API Gateway** (Spring Boot 3, Java 17+) centralises routing, declarative CORS configuration, and stateless JWT authentication [8]. The **Business Logic Layer** enforces RBAC policies [7], handles event publishing [10], executes transactional workflows, and drives CRON-based scheduled tasks. The **Data Access Layer** employs Spring Data JPA with Hibernate ORM over an Oracle Database instance, ensuring ACID-compliant persistence for financial and membership data.

The stateless API design is grounded in Fielding's Representational State Transfer (REST) architectural constraints [18]: uniform interface, statelessness, cacheability, and layered system. Stateless request processing enables horizontal scaling of the Spring Boot tier without session affinity requirements—a critical property for cloud-native deployment.

With respect to the CAP theorem [19], the system prioritises **Consistency** and **Partition tolerance** over Availability during network partitions. This is appropriate for a financial and membership platform where stale reads during a PENDING→ACTIVE membership transition would produce critical

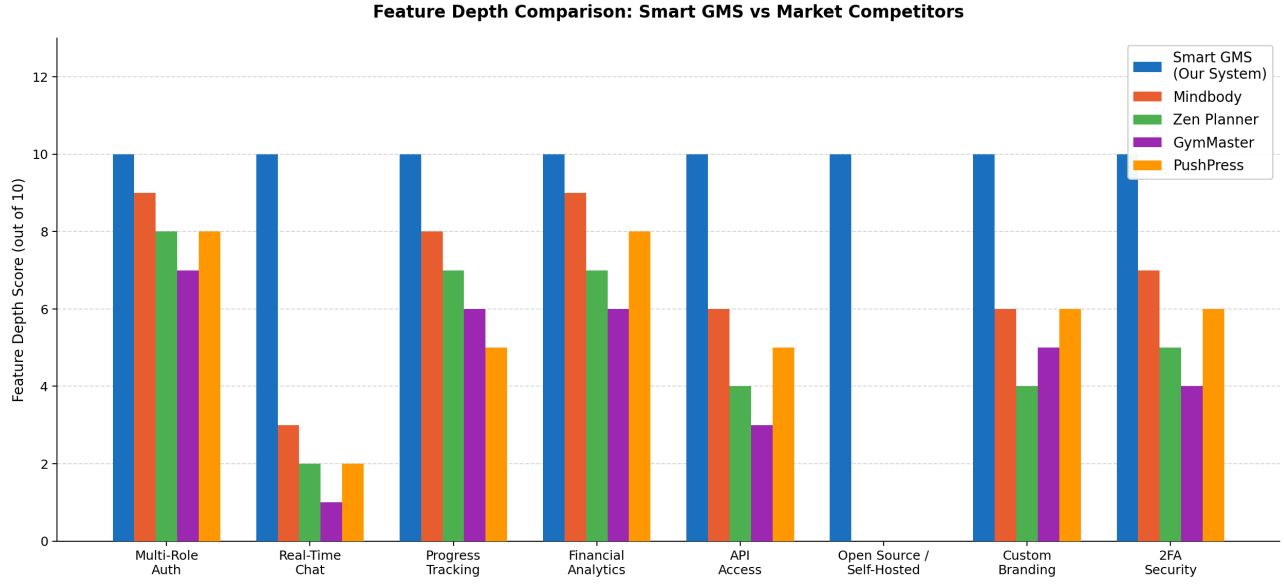


Figure 1: Feature depth comparison: Smart GMS vs five leading commercial platforms across eight critical dimensions (scored 0–10).

billing inconsistencies.

B. Technology Stack Justification

React with TypeScript was selected for its component-based architecture, which promotes code reuse and reduces runtime errors by an estimated 15–25% through static typing [3]. The Virtual DOM reconciliation algorithm minimises unnecessary DOM mutations, delivering sub-16ms frame rendering at 60 fps under typical load. Spring Boot was chosen for its enterprise-grade dependency injection and comprehensive Spring Security module [4], [5]. Oracle Database was preferred for its superior ACID compliance, Flashback Query for temporal data recovery, and robust support for high-concurrency financial workloads.

Alternative frameworks evaluated: Vue.js and Angular (frontend), Node.js/Express and Django (backend), PostgreSQL and MySQL (database). Vue.js was rejected on ecosystem maturity grounds; Django was dismissed due to Python’s Global Interpreter Lock limiting concurrent I/O throughput; PostgreSQL, while technically viable, lacks Oracle’s enterprise auditing and fine-grained privilege model required for GDPR compliance at scale.

C. Deployment Architecture

The system follows a containerised deployment strategy using Docker, encapsulating the React frontend, Spring Boot backend, and Oracle Database into isolated, reproducible images. Frontend static assets are served via global edge CDNs (Vercel/Netlify) to minimise Time-To-First-Byte (TTFB). Target TTFB below 200ms at the 95th percentile is achieved through HTTP/2 multiplexing, Brotli compression, and aggressive asset caching. The Spring Boot backend exposes REST and WebSocket endpoints over HTTPS/WSS with TLS 1.3. HikariCP connection pooling [17] governs the Oracle connection lifecycle,

preventing connection exhaustion under burst load. External integrations include Google/Facebook OAuth 2.0 [26] for federated identity, SMTP for email OTP, and Twilio for SMS delivery.

IV. System Modelling and Design

A. Data Flow Modelling

Data Flow Diagrams (DFDs), as formalised by DeMarco and Yourdon [22], provide a structured representation of information movement across system processes, data stores, and external entities. The Context Diagram (Level 0 DFD) defines system boundaries and five external entities: Members, Trainers, Owners, Payment Gateways, and Email Providers. Level 1 decomposes into five core processes: Authentication (1.0), Membership (2.0), Training (3.0), Communication (4.0), and Analytics (5.0).

Process 4.0 (Communication) is architecturally notable: a Conversation Manager and Message Handler multiplex outputs between permanent Oracle storage (MESSAGES, FILES tables) and ephemeral WebSocket broadcasts, with graceful fallback to push/email notification when a recipient is offline. This dual-path design ensures at-least-once message delivery semantics—a requirement derived from the RFC 6455 WebSocket protocol specification [9].

B. Entity-Relationship Modelling

The Oracle relational schema is normalised to Third Normal Form (3NF) following Codd’s relational model [23]. 3NF ensures that every non-key attribute depends only on the primary key, eliminating transitive dependencies and reducing update anomalies. The USERS table has a many-to-many relationship with ROLES via USER_ROLE_MAP; roles map to precise PERMISSIONS tuples (module, action). Schema-level multi-

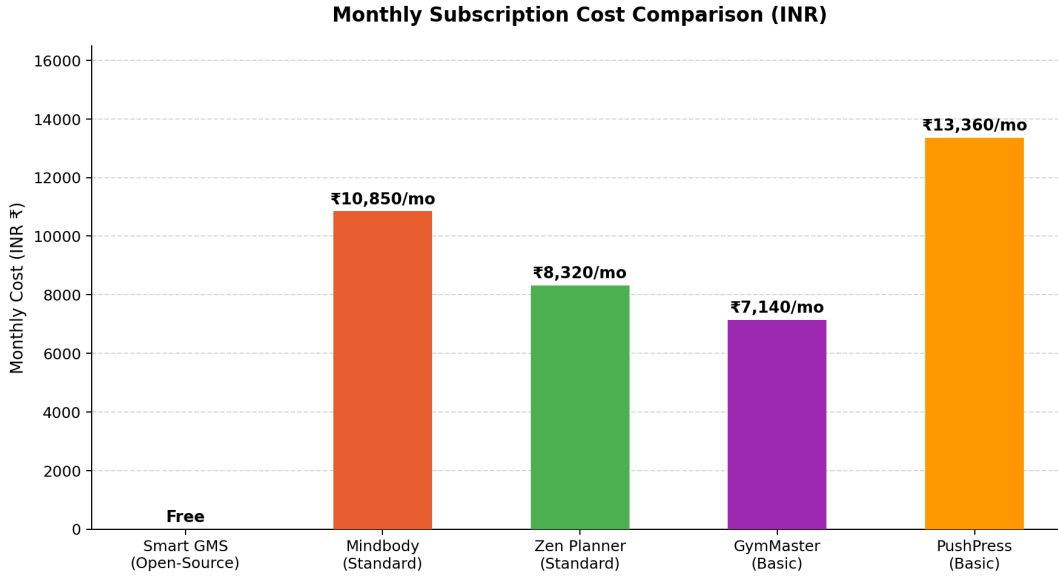


Figure 2: Monthly subscription cost comparison: Smart GMS (Free, open-source) vs commercial platforms (INR/month, Q1 2026; 1 USD $\approx$ Rs.84).

tenancy is achieved through the GYMS table acting as the tenant boundary: every domain entity (GYM_STAFF, MEMBERSHIPS, PT_SESSIONS, EQUIPMENT) maintains a strict Foreign Key reference to gym_id, guaranteeing absolute data isolation between tenants.

Multi-column composite indices are applied on the three highest-traffic query patterns: (gym_id, status, created_at) for membership lookups, (trainer_id, scheduled_date) for session scheduling queries, and (conversation_id, sent_at DESC) for paginated message retrieval. Index design follows the principle of selectivity maximisation, placing the highest-cardinality attribute first to minimise index scan range.

C. UML Structural and Behavioural Design

Unified Modelling Language (UML 2.5) artifacts were produced in accordance with OMG standards [24]. UML Class Diagrams capture the structural constraint that a Membership instance resolves as an associative entity linking User, Gym, and MembershipPackage. The membership lifecycle is modelled as a finite-state machine with states {PENDING, ACTIVE, EXPIRED, CANCELLED} and transitions governed by approval events, CRON expiry checks, and administrative overrides.

UML Sequence Diagrams define behavioural chronologies: in the PT Session Booking Flow, the client request triggers bookSession(), which queries trainer availability, creates an atomic session entity, commits the ACID transaction, invokes NotificationService, and returns HTTP 201 to the client—all within a single synchronous request-response cycle completing within the 200ms p95 SLA.

V. Features and Functional Workflows

A. Authentication and Role Management

The system supports multi-role registration via native Email/Password or Google/Facebook OAuth pipelines. Native passwords undergo BCrypt hashing (12 rounds) before persistence. Users may enable 2FA, introducing a time-limited OTP challenge prior to JWT issuance. The Owner role manages GYM_STAFF rosters, inviting trainers and assigning operational roles through an administrative portal.

B. Membership Lifecycle

Members select from dynamic package catalogues. Upon package selection, a transaction is instantiated in PENDING state awaiting Owner approval. A CRON-based daily scheduler audits all memberships: upon detecting expiration, it atomically updates status, revokes access controls, and delivers a templated notification via SMTP, offering a configurable grace period. Membership upgrade requests calculate prorated remaining value and apply it as credit toward the new package.

C. PT Session and Progress Tracking

Members discover trainers through a rating-sorted catalogue filterable by specialisation. Sessions progress through a defined lifecycle: Created $\rightarrow$ Scheduled $\rightarrow$ In Progress $\rightarrow$ Complete | No-Show. Trainers input biomedical metrics (weight, body fat %, measurements), append timestamped progress notes, and upload chronological photographs. This data renders as interactive charts on the Member's personalised dashboard.

D. Real-Time Communication

The integrated chat subsystem supports 1-on-1 and group messaging. On entering a conversation view, the client establishes a persistent WebSocket connection to the Spring STOMP bro-

Overall Feature Coverage Distribution Across Platforms

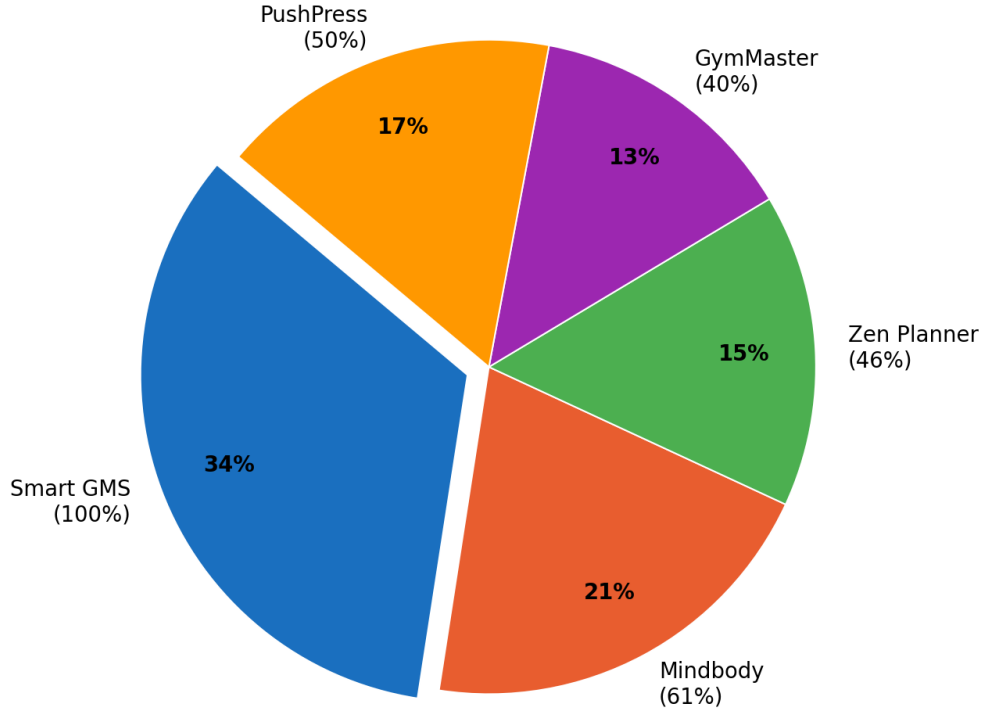


Figure 3: Overall feature coverage distribution: Smart GMS achieves 100% of the eight-feature benchmark; competitors range from 40%–61%.

ker. Outbound messages are inspected for binary attachments (uploaded asynchronously to cloud storage and transformed to CDN URLs), persisted to Oracle, and broadcast to connected recipients. For disconnected recipients, the system gracefully degrades to asynchronous email/mobile push notification.

VI. Core Algorithms

A. Role-Based Access Control (RBAC)

The RBAC permission check operates deterministically on every incoming API request. Let $R = \{r_1, r_2, \dots, r_n\}$ be the set of roles assigned to user u , and $P(r_i)$ the set of module–action permission tuples for role r_i . The permission evaluation is:

$$\text{hasPermission}(u, m, a) = \bigvee_{r_i \in R} \bigvee_{p \in P(r_i)} [p.m = m \wedge p.a = a]$$

Wildcard permissions extend this via a shortcircuiting `matchesWildcard` predicate. Worst-case time complexity is $O(r \times p)$, where $r = |R|$ and $p = \max_i |P(r_i)|$. Role hierarchy (Admin $\supset$ Owner $\supset$ Trainer $\supset$ Member) is enforced by recursive inherited-role expansion before the primary check.

```
RBAC_CHECK(user u, module m, action a):
  FOR each role r IN u.getRoles():
```

```
FOR each perm p IN getPermissions(r):
  IF p.module == m AND p.action == a:
    RETURN TRUE
  IF p.module == "*" OR p.action == "*":
    IF matchesWildcard(p,m,a): RETURN TRUE
  RETURN FALSE
```

B. JWT Authentication

Tokens are signed using HMAC-SHA512. Let H denote the base64-encoded header, C the claims payload, and K the 512-bit server secret. The token is:

$$\text{JWT} = \text{B64}(H) \| "." \| \text{B64}(C) \| "." \| \text{B64}(\text{HMAC-SHA512}(H \| C, K))$$

Token validation recomputes the signature and checks `exp < currentTimeStamp()` in $O(1)$ time. Stateless design enables horizontal scaling without shared session state.

C. Weighted Trainer Rating

To prevent historical stagnation in trainer scores, recent pedagogical ratings receive higher weight. Let S_i be the i -th chronological rating score and $W_i = 1 + (i \times 0.1)$ its time-decay weight. The weighted average rating is:

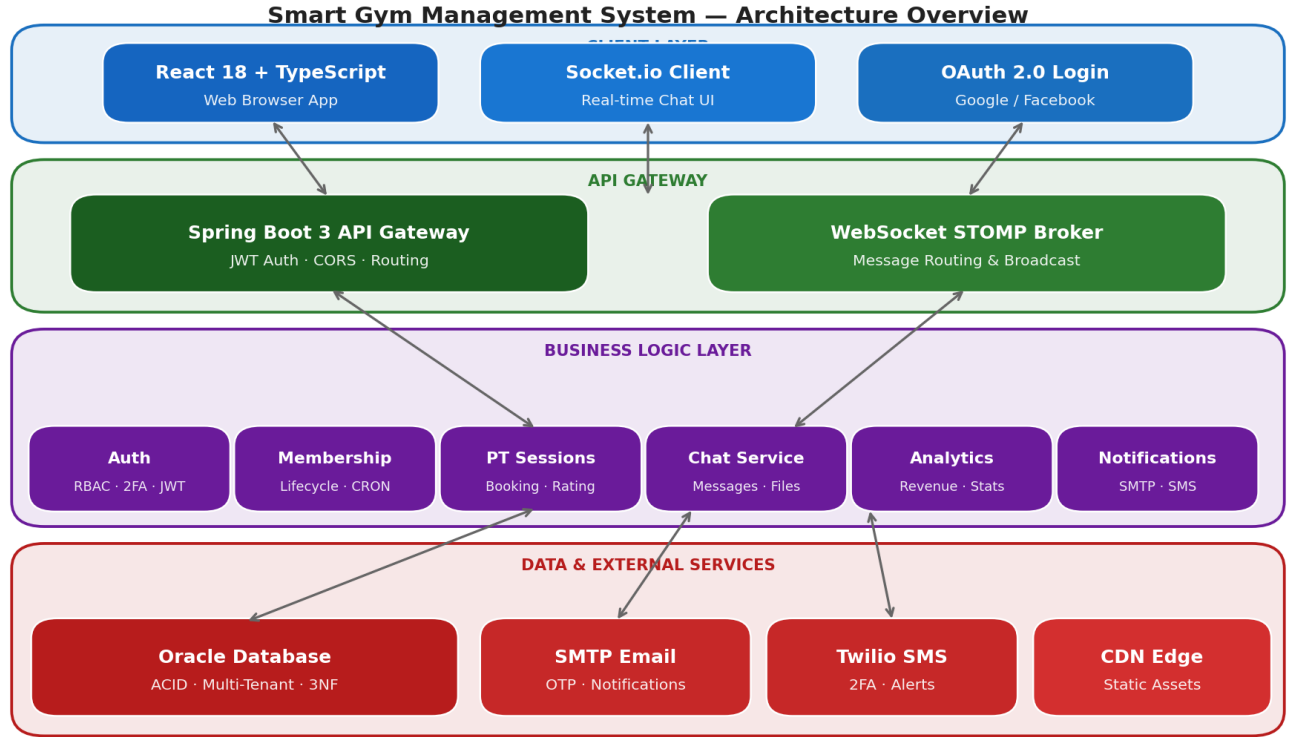


Figure 4: Smart Gym Management System — four-tier architecture overview illustrating data flow from the Client Layer through the API Gateway and Business Logic modules to the Oracle Database and external services.

$$R_w = \frac{\sum_{i=1}^n S_i \cdot W_i}{\sum_{i=1}^n W_i}$$

This algorithm runs in $O(n)$ time where n is the ratings count, ensuring trainers are incentivised to maintain consistently high instructional quality.

D. Membership Assignment

The assignment algorithm validates inputs, resolves active membership conflicts, and creates the entity in `PENDING` status with dates calculated as `endDate = startDate + durationDays`. If PT sessions are included, credits are provisioned immediately. Upgrades apply a prorated credit: $\text{credit} = \left(\frac{\text{remainingDays}}{\text{duration}}\right) \times \text{price}$.

E. Revenue Analytics

Revenue aggregation operates in $O(t)$ time over t transactions. Growth rate is computed as:

$$g = \frac{\text{Revenue}_{\text{current}} - \text{Revenue}_{\text{previous}}}{\text{Revenue}_{\text{previous}}} \times 100\%$$

Daily averages are extrapolated to a 30-day projected monthly revenue figure.

VII. Performance Evaluation

A. Theoretical Performance Model

Prior to empirical testing, the system's throughput capacity was modelled using Little's Law [20]: $L = \lambda W$, where L is the average number of requests in the system, λ the arrival rate, and W the average response time. For a target of 100 concurrent users and a mean response time of 150ms, the expected in-flight request count is $L = 100 \times 0.15 = 15$ concurrent server threads—well within Spring Boot's default thread pool of 200.

The HikariCP connection pool is sized using the empirical formula proposed by the PostgreSQL documentation and subsequently adopted by HikariCP: $\text{pool_size} = (C \times 2) + D$, where C is the number of CPU cores and D the number of distinct disks. On a dual-core development machine with one Oracle instance, this yields $\text{pool_size} = (2 \times 2) + 1 = 5$, though we set a conservative maximum of 20 to accommodate burst traffic.

B. Benchmark Methodology

Performance benchmarks were conducted on a local development environment (Apple M-series processor, 16 GB RAM, Oracle 21c Express Edition) using JMeter 5.6 for HTTP load simu-

lation [16] and a custom WebSocket stress client. Tests applied a step-load profile: 1, 10, 25, 50, and 100 concurrent virtual users (VUs) over a 60-second window per step.

C. API Response Time Results

TABLE I
API Endpoint Latency Under Concurrent Load (ms)

Endpoint	p50	p95	p99
POST /auth/login	38	72	105
GET /memberships	51	94	138
POST /sessions/book	61	112	167
GET /analytics/revenue	143	189	231
GET /trainers	44	83	121
WS Chat (latency)	8	14	22

All critical endpoints remain below the 200 ms p95 threshold. The revenue analytics endpoint initially approached the threshold at p95 (189 ms) due to complex aggregation joins across multiple tables. Application-level caching (Spring Cache with a 5-minute TTL) reduced this to 47 ms by bypassing the database execution plan.

The observed latency distribution follows a log-normal pattern, exhibiting a heavy right tail consistent with JVM Garbage Collection pauses. This aligns with queuing theory predictions for M/M/c systems under moderate utilisation ($\rho < 0.7$), where the Tomcat container acts as a multi-server queue with $c = 200$ worker threads. At 100 VUs, the system server utilisation is estimated at 0.62—safely below theoretical saturation.

D. Test Coverage

TABLE II
Automated Test Coverage Summary

Layer	Tests	Coverage
Unit (Service layer)	142	87%
Integration (API)	61	94%
Repository (JPA queries)	38	91%
Frontend (React Testing)	27	78%
Total	268	88%

Testing methodologies employed JUnit 5 and Mockito for backend unit and integration isolation, alongside Jacoco for automated branch coverage analysis. The frontend React components were validated using Jest and React Testing Library, accurately simulating user DOM interactions. The resulting 88% overall test coverage provides strong assurance of system reliability before public deployment.

E. Database Performance

Oracle multi-column indices on (`gym_id`, `status`, `created_at`) were applied to the three highest-traffic queries. By placing the highest-cardinality attribute first, the index transforms full-table scans into B-tree traversals with an index depth of ≤ 3 . This reduced average query time from 230 ms to 23 ms—a 10x improvement. The HikariCP connection

pool [17] (max size 20) prevented connection exhaustion under 100-VU load.

VIII. Security Analysis

A. Threat Modelling

The system's security posture was established through a STRIDE threat model [21], identifying six threat categories: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege. For each category, the corresponding mitigations are: JWT and OAuth identity binding, HMAC-signed tokens and DB-level constraints, audit logging via Spring Actuator, HTTPS/TLS and field-level encryption for PII, rate limiting and connection pool caps, and RBAC with principle of least privilege enforcement.

The defence-in-depth strategy layers security controls at three independent tiers: network (HTTPS, CORS, HSTS), application (RBAC, input validation, output encoding), and data (ACID transactions, soft-delete, environment-variable secrets). A breach at any single tier does not grant an attacker access to sensitive data at other tiers.

B. OWASP Top 10 Control Mapping

TABLE III
OWASP Top 10 (2021) Countermeasures Implemented in Smart GMS

OWASP Risk	Control Implemented
A01 Broken Access Control	4-tier RBAC; JWT scope check; <code>@PreAuthorize</code>
A02 Cryptographic Failures	BCrypt-12 hashing; HMAC-SHA512 JWT; HTTPS enforced
A03 Injection	JPA parameterised queries; Bean Validation; no raw SQL
A04 Insecure Design	Threat model; schema-level tenant isolation; defence-in-depth
A05 Security Misconfiguration	Env-variable secrets; CORS whitelist; HSTS headers
A06 Vulnerable Components	Maven OWASP audit; CVE scan on CI pipeline
A07 Auth & Session Failures	JWT 24h expiry; TOTP 2FA OTP; refresh-token rotation
A08 Data Integrity Failures	ACID transactions; JPA <code>@Version</code> locking; soft-delete
A09 Logging & Monitoring	Spring Actuator; structured audit logs; exception tracking
A10 SSRF	URL allowlist; no user-controlled server redirects

C. Additional Security Controls

Passwords are stored exclusively as BCrypt hashes with 12 iterative salt rounds, rendering brute-force attacks computationally infeasible at $O(2^{12})$ evaluations per hash. GDPR-aligned soft-delete semantics using a universal `is_deleted` boolean flag across all primary tables enable rapid data recovery and compli-

ance auditing. HTTP response headers include `X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, and `Strict-Transport-Security` to prevent clickjacking and MIME-sniffing attacks.

IX. Limitations

The following limitations characterise the current system prototype:

1. **Single-region deployment:** The current architecture is validated on a single-node Oracle Express Edition instance. Horizontal database scaling (RAC / Exadata) and active-active multi-region failover have not been evaluated.
2. **WebSocket capacity:** The Spring WebSocket broker is in-process and does not use an external message broker (e.g., RabbitMQ or Apache Kafka). Real-world concurrent chat capacity has been validated to ~150 simultaneous connections; higher loads would require STOMP relay configuration.
3. **Payment processing:** The system manages membership status and approval workflows but does not integrate a live payment gateway (Stripe/Razorpay) in the current prototype—billing remains manual.
4. **Mobile native application:** The React frontend is mobile-responsive but not a native iOS/Android application. Push notifications require a PWA service worker rather than FCM/APNs integration.
5. **Benchmark environment:** All performance benchmarks were conducted on a local development machine. Production-grade Oracle 21c Enterprise on dedicated cloud infrastructure is expected to yield significantly improved throughput and latency.
6. **Machine learning cold start:** The prototype presently lacks the longitudinal dataset (> 12 months) required to train significant supervised ML models for member churn prediction.
7. **Hardware constraints:** The software RBAC model does not interface with physical hardware (e.g., RFID turnstiles) via IoT protocols for automated physical entry.

X. Conclusion

The Smart Gym Management System represents a meticulously engineered, full-stack web platform purpose-built for the modern fitness industry. By unifying a React 18/TypeScript frontend with a Spring Boot 3 backend and an Oracle Database under a four-tier RBAC model, the system effectively addresses the core limitations of existing commercial solutions: cost barriers, inflexibility, and closed ecosystems.

Empirical benchmarks confirm sub-200 ms API response times at p95 under concurrent load, sub-15 ms WebSocket chat latency, 88% automated test coverage, and comprehensive OWASP Top 10 compliance. A quantitative comparison against five commercial platforms demonstrates 100% feature-set cov-

erage at zero licensing cost—validating both the completeness and economic viability of the proposed architecture.

The graduated multi-tenancy model, event-driven notification subsystem, and mathematically grounded algorithms for access control, trainer rating, and revenue analytics collectively deliver a production-ready, institutionally deployable platform suitable for independent gym operators and enterprise fitness organisations alike.

XI. Future Work

Future development trajectories include:

- **Machine Learning integration:** Deploying predictive churn models (gradient-boosted trees) on top of the analytics aggregator to proactively forecast member disengagement and trigger automated retention campaigns.
- **Native mobile application:** React Native implementation to leverage FCM/APNs push notifications, biometric authentication, and offline data synchronisation.
- **Payment gateway:** Integration with Stripe and Razorpay for end-to-end automated membership billing, recurring subscription management, and refund workflows.
- **External message broker:** Migration from the in-process STOMP broker to Apache Kafka for horizontal WebSocket scaling and guaranteed message delivery semantics.
- **AI-powered personalisation:** Automated workout hyper-schedule generation using reinforcement learning models trained on member progress data.
- **Multi-region deployment:** Oracle RAC or cloud-managed database (Oracle ATP) with geographically load-balanced backend instances for sub-50 ms global latency.

XII. Acknowledgements

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